

VelaDial — Hardware Validation Results

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Status: Active validation log — fill in incrementally as physical validation phases execute.

Project: VelaDial

Scope: Single consolidated results document, as defined by [docs/13_Firmware_Prep_Validation_Plan.md](#) §7.

How to use this document

- One section per validation phase (per [docs/13_Firmware_Prep_Validation_Plan.md](#) §3).
- Each section is owned by a named tester for a specific board / device in hand.
- Do **not** invent results. Any field whose value has not been physically observed must be recorded as [NOT TESTED — awaiting Hardik physical validation input](#).
- Do **not** edit [hardware/elecrow_pinout.md](#) based on guesses. Only update the pinout doc when physical deltas have been confirmed by Hardik on the actual board in hand.

Step 15B — ELECROW Door-Side Board / Pinout Validation Results

Implements the validation defined in [docs/validation/step_15a_elecrow_pinout_validation.md](#) (the Step 15A plan).

Identification

FIELD	VALUE
Board (model)	ELECROW CrowPanel 1.28 in ESP32-S3 round rotary touch display
Board revision / silkscreen / sticker	NOT TESTED — awaiting Hardik physical validation input
Validation date	NOT TESTED — awaiting Hardik physical validation input

FIELD	VALUE
Tester name	NOT TESTED — awaiting Hardik physical validation input
Bring-up YAML used	<code>esphome/door_side_rotary.yaml</code> (existing bring-up, unchanged)
Pinout source-of-truth used	<code>hardware/elecrow_pinout.md</code>

Evidence checklist

EVIDENCE ITEM	STATUS	LOCATION / REFERENCE
Board front photo captured	NOT TESTED — awaiting Hardik physical validation input	—
Board back photo captured	NOT TESTED — awaiting Hardik physical validation input	—
Revision marking (silkscreen / sticker) photo captured	NOT TESTED — awaiting Hardik physical validation input	—
Serial logs from bring-up flash captured	NOT TESTED — awaiting Hardik physical validation input	—
I ² C scan output captured	NOT TESTED — awaiting Hardik physical validation input	—

Photos and logs will be attached to the Step 15B PR (or stored in an agreed-upon location) once Hardik provides them. Do not commit raw binary assets to the repo unless the team explicitly decides to.

Subsystem results

Statuses: **PASS** / **PARTIAL** / **FAIL** / **BLOCKED** / **NOT TESTED**. (**PARTIAL** is only acceptable with a documented workaround approved by Hardik — see the Step 15B gate below.)

#	CHECK	EXPECTED RESULT	ACTUAL RESULT	STATUS	EVIDENCE / NOTES
1	Power-on baseline	Board powers up on USB 5 V / 2 A; power LED on GPIO40 behaves as	—	NOT TESTED — awaiting Hardik	—

#	CHECK	EXPECTED RESULT	ACTUAL RESULT	STATUS	EVIDENCE / NOTES
		documented; no brownout, no overheat, no reboot loop.		physical validation input	
2	Compile <code>esphome/door_side_rotary.yaml</code>	ESPHome compile completes without errors for the existing bring-up YAML, no pin / component conflicts.	—	NOT TESTED — awaiting Hardik physical validation input	Compile only — no behavior changes to YAML.
3	Flash bring-up YAML	Flash succeeds; device boots; serial log shows clean boot, no panic, no strap-pin warning, no reboot loop.	—	NOT TESTED — awaiting Hardik physical validation input	—
4	Display — GC9A01A	Display initializes over SPI on the documented pins (SCLK 10, MOSI 11, DC 3, CS 9, RST 14); LVGL bring-up page renders ("VelaDial" / "Press dial").	—	NOT TESTED — awaiting Hardik physical validation input	Confirm <code>invert_colors: true</code> is still correct for this panel; record if not.
5	Backlight PWM	Backlight on GPIO46 sweeps smoothly 0 % → 100 % via the <code>back_light</code> light entity; visible change; no flicker / whine.	—	NOT TESTED — awaiting Hardik physical validation input	—
6	CST816 I ² C address <code>0x15</code>	I ² C scan on SDA GPIO6 / SCL GPIO7 reports a device at <code>0x15</code> .	—	NOT TESTED — awaiting Hardik	—

#	CHECK	EXPECTED RESULT	ACTUAL RESULT	STATUS	EVIDENCE / NOTES
				physical validation input	
7	CST816 coordinate orientation	Touches generate coordinates; existing transform (<code>mirror_y: true</code> , <code>swap_xy: true</code>) maps touches to visible panel correctly.	—	NOT TESTED — awaiting Hardik physical validation input	Record any required transform change — do not "fix" in production YAML in Step 15B.
8	Rotary encoder CW / CCW	Rotating CW and CCW changes encoder count in opposite directions on pins A=GPIO45, B=GPIO42.	—	NOT TESTED — awaiting Hardik physical validation input	Record observed direction. Do not "fix" direction in production YAML in Step 15B.
9	Rotary encoder press	Pressing the encoder shaft registers a press on GPIO41 (<code>rotary_button</code> in bring-up YAML).	—	NOT TESTED — awaiting Hardik physical validation input	—
10	WS2812 LED ring (5 LEDs)	All 5 LEDs on GPIO48 light up in order from a temporary test entity; no dropped pixels.	—	NOT TESTED — awaiting Hardik physical validation input	Use a temporary test entity only — do not commit it to production YAML.
11	30-minute stability soak	Board runs ≥ 30 min with display + backlight + touch active; no reboots, no I ² C bus errors, no thermal issues.	—	NOT TESTED — awaiting Hardik physical	Attach final serial-log tail to evidence.

#	CHECK	EXPECTED RESULT	ACTUAL RESULT	STATUS	EVIDENCE / NOTES
				validation input	

Pinout delta vs hardware/elecrow_pinout.md

Confirmed matches

SUBSYSTEM	FUNCTION	DOCUMENTED (HARDWARE/ ELECROW_PINOUT.MD)	PHYSICALLY CONFIRMED?
Display	SCLK	GPIO10	NOT TESTED — awaiting Hardik physical validation input
Display	MOSI	GPIO11	NOT TESTED — awaiting Hardik physical validation input
Display	DC	GPIO3	NOT TESTED — awaiting Hardik physical validation input
Display	CS	GPIO9	NOT TESTED — awaiting Hardik physical validation input
Display	RST	GPIO14	NOT TESTED — awaiting Hardik physical validation input
Display	Backlight (PWM)	GPIO46	NOT TESTED — awaiting Hardik physical validation input
Touch	SDA	GPIO6	NOT TESTED — awaiting Hardik physical validation input
Touch	SCL	GPIO7	NOT TESTED — awaiting Hardik physical validation input
Touch	INT	GPIO5	NOT TESTED — awaiting Hardik physical validation input
Touch	RST	GPIO13	NOT TESTED — awaiting Hardik physical validation input
Touch	I ² C address 0x15	0x15	NOT TESTED — awaiting Hardik physical validation input
Encoder	A	GPIO45	

SUBSYSTEM	FUNCTION	DOCUMENTED (HARDWARE/ELECROW_PINOUT.MD)	PHYSICALLY CONFIRMED?
			NOT TESTED — awaiting Hardik physical validation input
Encoder	B	GPIO42	NOT TESTED — awaiting Hardik physical validation input
Encoder	Switch	GPIO41	NOT TESTED — awaiting Hardik physical validation input
WS2812	Data	GPIO48	NOT TESTED — awaiting Hardik physical validation input
WS2812	LED count	5	NOT TESTED — awaiting Hardik physical validation input
Power indicator	Power LED	GPIO40	NOT TESTED — awaiting Hardik physical validation input

Confirmed mismatches

SUBSYSTEM	FUNCTION	DOCUMENTED	OBSERVED ON BOARD	NOTES / PROPOSED ACTION
<i>none yet</i>	—	—	—	NOT TESTED — awaiting Hardik physical validation input

 Do **not** edit [hardware/elecrow_pinout.md](#) based on entries here until Hardik has physically confirmed the delta and explicitly approved the pinout-doc update in a separate, narrowly-scoped PR.

Unresolved TBDs

ITEM	WHY IT IS TBD	RESOLUTION PATH
Board revision / silkscreen	Not yet observed in hand	Photograph silkscreen, record value here.
Onboard OLED presence (SDA GPIO38 , SCL GPIO39)	Optional on some board revisions	Visual inspection + I ² C scan on bus, record presence/absence.

ITEM	WHY IT IS TBD	RESOLUTION PATH
Test IO usage (<code>GPI04</code> , <code>GPI012</code>)	Listed as test IO in pinout doc; intended door-side function not finalized	Out of scope for Step 15B unless they conflict with bring-up.

Step 15B Gate

 *Do not proceed to production door-side firmware, production door-side YAML, the 3-page LVGL UI, door-side sensor wiring (TSL2591 / SHT45), Home Assistant action wiring, VL53L4CD work, or any sensor fusion work until Step 15B is completed end-to-end and the results in this document have been reviewed and approved by Hardik.*

Specifically, until this gate is lifted, do **not**:

- Modify `esphome/door_side_rotary.yaml` for production behavior.
- Create any new production YAML under `esphome/`.
- Begin door-side sensor (TSL2591, SHT45) validation (`docs/13_Firmware_Prep_Validation_Plan.md` §3.C).
- Begin VL53L4CD / VL53L0X work (`docs/13_Firmware_Prep_Validation_Plan.md` §3.A / §3.E).
- Begin LVGL Power / Brightness / Presets page implementation.
- Begin sensor fusion work (explicitly out of v1).

Gate lift requires: all subsystem rows above marked **PASS** (or **PARTIAL** with documented workaround approved by Hardik), all evidence items captured, and any pinout deltas resolved.

Step 15B gate — process waiver note (owner-acknowledged)

 **Process waiver acknowledged by Hardik (owner).**

PR #19 created a production-oriented door-side YAML (`esphome/door_side_rotary.yaml` , full LVGL 3-page UI + wake-only-first guards + preset/brightness/HA-state logic) **before** any physical Step 15B validation was completed. This is accepted **only** as a pre-hardware draft for compile-time and design-level review.

This acceptance does **NOT** lift the Step 15B physical validation gate. No physical PASS is claimed. All hardware rows in this document remain **NOT TESTED**. The YAML's own header continues to mark it `DRAFT – COMPILER PASSED / HARDWARE VALIDATION PENDING` .

The gate continues to apply to: - All further production-firmware-behavior changes to `esphome/door_side_rotary.yaml` (beyond compile/lint cleanups that do not change behavior). - Any new production YAML under `esphome/`. - Door-side sensor wiring (TSL2591 / SHT45) validation. - LVGL UI sign-off as production-ready. - VL53L4CD / VL53L0X / sensor fusion work (independently gated).

The gate is **lifted only** when Hardik records physical Step 15B PASS results (or owner-approved PARTIAL with workaround) in the subsystem table above, with evidence captured, and explicitly signs off in this section.

Phase 1 — Door-side ESP32-S3 Firmware Draft

```

**Date:** 2026-05-24
**Status:** Draft created (PR #19 merged)
**Compile status:** PASSED in Manus environment using ESPHome 2026.5.0 (binary size 1,194,131 bytes, 0 errors, 0 warnings, 181.5s)

A draft production-oriented YAML (`esphome/door_side_rotary.yaml`) has been created. The YAML compiled without errors.

## Phase 2 — Bedside ESP32-C6 APDS Firmware Draft

**Date:** 2026-05-24
**Status:** Draft created
**Compile status:** PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 0 warnings, 181.5s)
**Physical Validation:** NOT TESTED

A draft production-oriented YAML (`esphome/bedside_gesture.yaml`) has been created on branch `firmware/bedside-gesture`

```

Pin Cross-Validation (Source Research)

The YAML pin assignments were cross-validated against **three independent sources** on 2026-05-24:

SOURCE	URL	AGREEMENT
Elecrow Official GitHub (factory source code)	https://github.com/Elecrow-RD/CrowPanel-1.28inch-HMI-ESP32-Rotary-Display-240-240-IPS-Round-Touch-Knob-Screen	Full match
Incipiens Community ESPHome YAML (XDA/HA)	https://github.com/Incipiens/Elecrow-Rotary-Displays	Full match
Makerguides.com Tutorial (pin table)	https://www.makerguides.com/getting-started-crowpanel-1-28inch-hmi-esp32-rotary-display/	Full match

All three sources confirm: Display SPI (SCLK=10, MOSI=11, CS=9, DC=3, RST=14), Backlight=GPIO46, Touch (SDA=6, SCL=7, INT=5, RST=13, addr 0x15), Encoder (A=45, B=42, SW=41), WS2812 (GPIO48, 5 LEDs), Power LED (GPIO40), invert_colors=true, touch transform (mirror_y=true, swap_xy=true).

This does NOT replace physical validation. It confirms the YAML is aligned with all known documentation sources.

Phase 3 — Raspberry Pi / Home Assistant Setup Guide

Date: 2026-05-25

Status: Setup guide created ([docs/setup/raspberry_pi_home_assistant_setup.md](#))

HA / ESPHome / LocalTuya Setup: NOT TESTED

Physical PASS Results Added: No

A comprehensive setup guide has been created documenting the intended deployment path for Home Assistant OS on Raspberry Pi, ESPHome add-on configuration, LocalTuya integration, and [light.bedroom_group](#) creation. All steps remain NOT TESTED until Hardik performs the physical setup.

Phase 4 — Full E2E Setup and Validation Guide

Date: 2026-05-25

Status: E2E guide created ([docs/setup/full_e2e_setup_and_validation_guide.md](#))

E2E Validation: NOT TESTED

Physical PASS Results Added: No

A full E2E setup and validation guide has been created. It defines the complete validation sequence for door-side, bedside, HA/LocalTuya command path, and full system scenarios. All results remain NOT TESTED until Hardik performs them on physical hardware.

Phase 5 — UI/UX Guide Integration Plan

Date: 2026-05-25

Status: UI/UX integration plan created ([docs/ui/ux_guide_integration_plan.md](#))

UI Implementation: NOT PERFORMED

Hardik UI/UX Guide: PENDING HARDIK INPUT

Physical PASS Results Added: No

A UI/UX guide integration plan has been created documenting the agreed visual themes, locked v1 rules, pending inputs from Hardik, integration decision matrix, and acceptance criteria for future UI implementation. No UI code changes were made. No ESPHome YAML was edited. Physical validation remains NOT TESTED.

Phase 6 — Claude Review Package

Date: 2026-05-25

Status: Claude review package created ([docs/review/claude_review_package.md](#))

Validation Performed: No

Physical PASS Results Added: No

A Claude review package has been created for independent review of the VelaDial repository after Phases 0–5. This is a documentation-only deliverable. No validation was performed, no firmware was changed, and all hardware/E2E validation remains NOT TESTED.

UI Concept Direction Package Note

Date: 2026-05-26

Status: UI concept direction package created (6 documents in [docs/ui/](#))

Validation Performed: No

Physical PASS Results Added: No

A premium UI concept direction package has been created with 20 deeply expanded concepts, a scored shortlist, selection criteria with pass/fail gates, a creative design notebook, a visual batch plan, and implementation candidates. This is a documentation-only deliverable. No firmware was changed, no YAML was modified, and all hardware/E2E validation remains NOT TESTED. Hardik must select a concept direction before any UI implementation begins.

UI Concept 01: Minimal Thermostat Note

Date: 2026-05-27

Status: Concept prototype YAML created and compile PASSED (ESPHome 2026.5.0, 177s, 0 errors)

Validation Performed: No

Physical PASS Results Added: No

Concept 01 (Minimal Thermostat) has been prototyped as a single-page UI concept for the door-side ELECROW controller. The concept uses a central label for power/brightness display and a surrounding arc for brightness indication. Knob CW/CCW adjusts brightness, knob press toggles power. Wake-only-first logic is preserved. This is a concept exploration only — not production, not tested on physical hardware. Four documentation files (research.md, design_spec.md, implementation_notes.md, validation_notes.md) accompany the YAML.

UI Concept 02: SmartKnob-Inspired Arc Note

Date: 2026-05-27

Status: Concept prototype YAML created and compile PASSED (ESPHome 2026.5.0, 361s, 0 errors)

Validation Performed: No

Physical PASS Results Added: No

Concept 02 (SmartKnob-Inspired Arc) has been prototyped as a single-page UI concept for the door-side ELECROW controller. The concept features a prominent SmartKnob-style arc widget spanning 270 degrees (135° to 45°) with an amber indicator and white knob handle for brightness visualization. The arc is visual-only (no touch drag) — all interaction flows through the physical rotary encoder (CW/CCW for brightness, press for power toggle). A large central percentage label doubles as a touch-to-toggle power button. Wake-only-first logic is enforced on all input paths (touch, knob CW/CCW, knob press). Home Assistant state import keeps the UI synchronized with `light.bedroom_group`. This is a concept exploration only — not production, not tested on physical hardware. Four documentation files (research.md, design_spec.md, implementation_notes.md, validation_notes.md) accompany the YAML.

UI Concept 03: Large Center Power Button Note

FIELD	VALUE
Concept YAML	<code>esphome/concepts/door_side_concept_03_large_center_power.yaml</code>

FIELD	VALUE
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 386s)
Physical validation	NOT TESTED — awaiting Hardik physical validation input

Concept 03 (Large Center Power Button) has been prototyped as a 3-page UI concept for the door-side ELECROW controller. The concept features an oversized 140px glowing circular power button as the hero element on the Power page, with a brightness arc on Page 2 and a diamond-layout preset selector on Page 3. The power button uses LVGL shadow properties to create an amber glow halo when on, and a ghost outline when off. Wake-only-first logic is enforced on all input paths (touch, knob CW/CCW, knob press). LED ring mirrors the power state (amber when on, off when off). Home Assistant state import keeps the UI synchronized with `light.bedroom_group`. This is a concept exploration only — not production, not tested on physical hardware. Four documentation files (research.md, design_spec.md, implementation_notes.md, validation_notes.md) accompany the YAML.

UI Concept 04: Single-Page Simple Mode Note

FIELD	VALUE
Concept YAML	<code>esphome/concepts/door_side_concept_04_single_page_simple.yaml</code>
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 374s)
Gate G1 status	FAILS — uses 1 page instead of mandated 3
Physical validation	NOT TESTED — awaiting Hardik physical validation input

Concept 04 (Single-Page Simple Mode) has been prototyped as a single-page UI concept for the door-side ELECROW controller. The concept places ALL controls on one screen: power state in the top band (tap to toggle), brightness percentage with thin arc in the center band (knob-only control), and active preset name in the bottom band (tap to cycle through 4 presets). This is the most radical simplification in the 20-concept matrix — no page navigation, no swipe gestures, no page indicator dots. Wake-only-first logic is enforced on all input paths. **This concept FAILS Gate G1 (Three-Page Lock) and cannot be selected for production without explicit owner waiver.** This is a concept exploration only — not production, not tested on physical hardware.

UI Concept 05: Preset Ring UI Note

FIELD	VALUE
Concept YAML	<code>esphome/concepts/door_side_concept_05_preset_ring.yaml</code>
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 378s)
Gate status	PASSES all gates (G1-G8)
Physical validation	NOT TESTED

Concept 05 (Preset Ring UI) uses 4 colored arc segments around the ring as a preset selector on the Presets page. The ring metaphor unifies all 3 pages: full ring (power), proportional arc (brightness), segmented ring (presets). Knob rotation cycles through presets; press applies. LED ring mirrors preset color. Wake-only-first enforced.

UI Concept 06: Night Mode Ultra-Minimal Note

FIELD	VALUE
Concept YAML	<code>esphome/concepts/door_side_concept_06_night_mode_ultra_minimal.yaml</code>
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 86s incremental)
Gate status	PASSES all gates (G1-G8) — Layer concept
Physical validation	NOT TESTED

Concept 06 (Night Mode Ultra-Minimal) is a layer concept — the sleep/dark-room state that overlays any active primary concept. When the TSL2591 ambient light sensor reads below 5 lux for 10 seconds, the display enters Night Mode: backlight drops to 10% PWM, all UI elements disappear except a single 8px amber dot (lights off) or dim percentage value (lights on). ALL inputs in Night Mode are wake-only — no action is executed, only the full UI is restored. Hysteresis band (5 lux entry, 15 lux exit) prevents oscillation.

UI Concept 07: Text-First Utility Note

FIELD	VALUE
Concept YAML	<code>esphome/concepts/door_side_concept_07_text_first_utility.yaml</code>
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 464s)
Gate status	PASSES all gates (G1-G8) — v1 candidate
Physical validation	NOT TESTED

Concept 07 (Text-First Utility) rejects all decorative elements. No arcs, rings, gradients, or glow effects. Every piece of information is communicated through typography alone: 56pt primary values (ON/OFF/percentage), 24pt preset names in a vertical list, 16pt section headers. The amber accent color is used only for active state and page dots. Design reference: Massimo Vignelli subway signage, aviation instrumentation. Best daylight readability of any concept in the matrix.

UI Concept 08: Apple Watch Complications Note

FIELD	VALUE
Concept YAML	<code>esphome/concepts/door_side_concept_08_apple_watch_complications.yaml</code>
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 326s)
Gate status	PASSES all gates (G1-G8) — v1-expanded adaptation only
v1 Recommendation	NOT RECOMMENDED per direction matrix — too complex
Physical validation	NOT TESTED

Concept 08 (Apple Watch Complications) borrows the watchOS Infograph "complications" metaphor: small data widgets at the top corners of each page showing ambient sensor data (WiFi RSSI top-left, ambient lux top-right). The v1-expanded adaptation uses only 2 complications per page in static gray, display-only (not interactive). The center content remains the hero element on each page. At 190 PPI (vs Apple Watch 326 PPI), information density is deliberately reduced. Full complication face with 4-6 interactive complications is v2 scope only.

UI Concept 09: LED-Ring Status-First Note

FIELD	VALUE
Concept YAML	esphome/concepts/door_side_concept_09_led_ring_status_first.yaml
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 81s)
Gate status	PASSES all gates (G1-G8) — Layer concept
Physical validation	NOT TESTED
LED hardware tests	ALL PENDING (color order, LED 0 position, brightness ceiling, RMT conflicts)

Concept 09 (LED-Ring Status-First) inverts the typical smart-display hierarchy. The 5-LED WS2812 ring on GPIO48 is the **PRIMARY** output — visible from across the room — while the 240×240 screen is secondary with only 6 LVGL widgets (3 labels + 3 page dots). LED color maps to active preset (amber/gold/white/dim amber), LED brightness is proportional to light brightness (capped at 50% for bedroom safety), and error states use red pulse patterns. The LED ring has an extended 30-second timeout after screen sleep for ambient room awareness. All LED behavior is hardware-test-pending: color order (GRB assumed), LED 0 physical orientation, brightness ceiling comfort, and RMT channel conflicts are unverified.

UI Concept 10: Three-Screen Tab Carousel Note

FIELD	VALUE
Concept YAML	esphome/concepts/door_side_concept_10_three_screen_tab_carousel.yaml
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 417s)
Gate status	PASSES all gates (G1-G8) — Direct v1 spec implementation
Physical validation	NOT TESTED

Concept 10 (Three-Screen Tab Carousel) is the direct, literal implementation of the locked v1 specification. Three horizontally swipeable pages (Power, Brightness, Presets) with a 3-dot page indicator in the LVGL top_layer (always visible). The concept uses the production YAML's exact architecture: FLEX-layout dot container, guarded swipe scripts, 2×2 preset grid with FLEX ROW_WRAP, and a display-only brightness arc controlled exclusively by the rotary encoder. Premium

execution details: 200ms MOVE_LEFT/MOVE_RIGHT transitions, active dot grows from 8px to 10px, amber accent throughout. This is the safe baseline against which all other concepts are compared. Zero compile errors on first attempt.

UI Concept 11: Brightness-First UI Note

FIELD	VALUE
Concept YAML	<code>esphome/concepts/door_side_concept_11_brightness_first_ui.yaml</code>
Compile status	PASSED (ESPHome 2026.5.0, ESP-IDF, 0 errors, 429s)
Gate status	PASSES all gates (G1-G8) — IA innovation
Physical validation	NOT TESTED

Concept 11 (Brightness-First UI) is architecturally identical to Concept 10 with one critical information architecture change: the default landing page is Brightness (page 0) instead of Power. The hypothesis is that "adjust brightness" is the most common bedroom action — especially at night. Users wake the device and immediately see the current brightness level and can rotate the knob to adjust without any page navigation. Knob press on the Brightness page is a power toggle shortcut, making the two most common actions (adjust brightness, toggle power) accessible without any swipe. Zero compile errors on first attempt. Page order: Brightness → Power → Presets.

Concept 12: Door Switch Replacement

ITEM	STATUS
Compile (ESPHome 2026.5.0)	PASSED (0 errors, 446s)
Physical validation	NOT TESTED

Concept 12 (Door Switch Replacement) treats the device as a premium replacement for a bedroom wall switch. The Power page is the hero — a 230×230px circular touch target fills the entire display. ON state floods the screen with amber (0xFFA500), OFF shows a dark circle with a thin white border. The mental model is: "the entire device is the switch." Visible from the doorway — amber glow = lights on, dark = lights off. Knob press = power toggle on ALL pages (not just Power page). Guest-friendly: zero learning curve, no hidden gestures needed. LED ring at 50% max brightness for door-side visibility. Context-aware sleep: Power page sleeps after 30s (transient door-side use), other pages after 60s. First compile attempt failed due to transient network error downloading LVGL component; second attempt passed cleanly with 0 errors.

Concept 13: Lunar Phase Visualization

ITEM	STATUS
Compile (ESPHome 2026.5.0)	PASSED (0 errors, 38s incremental)
Physical validation	NOT TESTED

Concept 13 (Lunar Phase Visualization) uses a moon phase metaphor to represent brightness state. 100% brightness = full moon (amber disc fills the display), 0% = new moon (dark with a thin limb crescent). The terminator line is approximated using overlapping LVGL circles — a bright disc (the illuminated surface) and a dark disc (the shadow) whose horizontal offset maps to brightness percentage. This creates a waxing/waning visual as the knob rotates. The round display IS the moon — form factor and metaphor are unified. Page 1 (Power) shows full moon vs new moon. Page 2 (Brightness Hero) shows the phase with a semi-transparent percentage overlay. Page 3 (Presets) maps each preset to a named lunar phase: Warm White = Full Moon, Soft Amber = Waxing Gibbous, Neutral White = First Quarter, Low Nightlight = Crescent. LED ring mirrors moon state with proportional amber. Poetic and premium — designed for bedroom ambiance.

Concept 14: Sundial Shadow UI

ITEM	STATUS
Compile (ESPHome 2026.5.0)	PASSED (0 errors, 106s fresh build)
Physical validation	NOT TESTED

Concept 14 (Sundial Shadow UI) maps brightness to the length of a shadow cast by a virtual gnomon at the center of the round display. At 100% brightness, the shadow is short (high noon). At 5%, the shadow is long (twilight). The shadow is approximated using an LVGL arc widget — a dark wedge whose angular span grows inversely with brightness. The background color shifts from warm amber (noon) to charcoal (twilight). Page 1 (Power) shows sunrise (ON) vs sunset (OFF) via the sundial face color. Page 2 (Brightness Hero) shows the shadow arc with a percentage overlay. Page 3 (Presets) maps each preset to a sun position: Warm White = Noon, Soft Amber = Golden Hour, Neutral White = Morning, Low Nightlight = Twilight. LED ring acts as a sun halo with proportional warm amber. Premium and architectural — connects electric light to natural light.

Concept 15: Tree Ring Growth Pattern

ITEM	STATUS
Compile (ESPHome 2026.5.0)	PASSED (0 errors, 163s fresh build)

ITEM	STATUS
Physical validation	NOT TESTED

Concept 15 (Tree Ring Growth Pattern) maps brightness to the number of visible concentric rings on the display, like the cross-section of a tree trunk. At 100% brightness, the display is filled with 8 concentric rings extending to the bezel — a mature tree. At 5%, only the innermost heartwood is visible. The rings are implemented as concentric `lv_obj` circles with borders in a gradient of wood tones (dark heartwood to light amber sapwood). Each ring's border opacity is toggled based on brightness threshold. Page 1 (Power) shows warm heartwood (ON) vs dark dormant (OFF). Page 2 (Brightness Hero) shows the ring growth with a percentage overlay. Page 3 (Presets) maps each preset to a season: Warm White = Summer, Soft Amber = Autumn, Neutral White = Spring, Low Nightlight = Winter. LED ring acts as the outermost growth ring with proportional warm amber. Premium and organic — connects electric light to natural growth.

Concept 16: Topographic Contour Map

ITEM	STATUS
Compile (ESPHome 2026.5.0)	PASSED (0 errors, 159s fresh build)
Physical validation	NOT TESTED

Concept 16 (Topographic Contour Map) maps brightness to terrain elevation. At 100% brightness, the display shows a mountain peak — 12 dense, closely spaced contour lines filling the circle with a bright summit marker at center. At 5%, only the innermost contour is visible (valley floor). The contour lines are implemented as thin-bordered (2px) concentric `lv_obj` circles in a gradient from dark amber (inner) to bright amber (outer), creating a cartographic aesthetic. A small summit marker dot at center differentiates this from Concept 15 (Tree Rings). Page 1 (Power) shows active terrain (ON) vs flat plain (OFF). Page 2 (Brightness Hero) shows contour density with elevation percentage. Page 3 (Presets) maps each preset to a terrain zone: Warm White = Summit, Soft Amber = Ridge, Neutral White = Plateau, Low Nightlight = Valley. LED ring acts as the outer horizon line with proportional warm amber. Premium and cartographic — connects electric light to terrain exploration. Uses Roboto Mono for technical precision feel.

Concept 17: Iris Aperture Mechanism

ITEM	STATUS
Compile (ESPHome 2026.5.0)	PASSED (0 errors, 169.83s

ITEM	STATUS
	fresh build)
Physical validation	NOT TESTED

Concept 17 (Iris Aperture Mechanism) maps brightness to the opening diameter of a camera iris/aperture. At 100% brightness, the iris is fully open (f/1.4) — a large warm amber circle fills the display, representing maximum light entering the lens. At 5%, the iris is nearly closed (f/16) — only a tiny pinhole of amber light is visible through the converging blades. The iris blades are simulated using 8 thick `lv_arc` segments arranged in an octagonal pattern, representing the overlapping metallic blades of a precision camera diaphragm. The central amber circle (the "light field") scales proportionally with brightness. Page 1 (Power) shows an open iris (ON) vs fully closed iris (OFF). Page 2 (Brightness Hero) shows the iris opening with a percentage overlay. Page 3 (Presets) maps each preset to an aperture mode: Warm White = Wide Open, Soft Amber = Portrait, Neutral White = Landscape, Low Nightlight = Pinhole. LED ring acts as the lens rim / light leak with proportional warm amber. Premium and optical/mechanical — connects electric light to precision optics. The most mechanically coherent concept in the matrix: aperture literally controls light.

Concept 18: Radar Sweep Animation

Item	Status
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Compile (ESPHome 2026.5.0)	PASSED (0 errors, 150.57s fresh build)
Physical validation	NOT TESTED

Concept 18 (Radar Sweep Animation) maps brightness to the angular span of a radar sweep trail. At 100% brightness, the sweep arc forms a full 360° circle (maximum scan coverage). At 5%, only a tiny 18° sliver is visible (minimal scan). The display is styled as a futuristic scanning instrument with concentric range rings (dark gray circles at `0x222222`), faint crosshair lines, and a central scope dot. The sweep trail is implemented as a large `lv_arc` widget with a warm amber indicator (`0xFFB000`) whose `value` maps directly to brightness percentage. A secondary inner arc provides visual depth. Page 0 (Power) shows "SCAN ACTIVE" (ON, large amber blip) vs "SCAN OFFLINE" (OFF, dim gray dot). Page 1 (Brightness Hero) shows the sweep scope with range rings, dual sweep arcs, and a large Roboto Mono percentage readout. Page 2

ITEM	STATUS
(Presets) maps each preset to a scan mode: Warm White = Wide Scan, Soft Amber = Focused Scan, Neutral White = Daylight Scan, Low Nightlight = Stealth Scan. LED ring acts as scanner perimeter glow with proportional warm amber. Premium and futuristic — connects electric light to environmental scanning. The only concept in the matrix with a motion/animation metaphor. Uses arc trail approximation instead of true continuous rotation for compile safety and performance.	
### Concept 19: Vinyl DJ Crossfader	
Item	Status
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Compile (ESPHome 2026.5.0)	PASSED (0 errors, 147.68s fresh build)
Physical validation	NOT TESTED

Concept 19 (Vinyl DJ Crossfader) maps brightness to the radial position of a virtual "needle" on a vinyl record. At 100% brightness, the needle sits on the outermost track band (full volume/brightness). At 5%, the needle sits on the innermost track band near the center label. The display is styled as a premium turntable platter viewed from above with four concentric "track bands" (14px-wide `lv_arc` widgets) representing different brightness zones. The active band illuminates in warm amber (`0xFFB000`) while inactive bands remain dark gray (`0x222222`). A central "record label" area (60px circle) displays the brightness percentage. Page 0 (Power) shows "SPINNING" (ON, amber spindle) vs "STOPPED" (OFF, gray spindle) with decorative groove rings. Page 1 (Brightness Hero) shows four track bands with the active one highlighted based on brightness zone. Page 2 (Presets) maps each preset to a "track" on the record: Track A (outer) = Warm White, Track B = Soft Amber, Track C = Neutral White, Track D (inner) = Low Nightlight. Tapping a track zone "drops the needle" on that preset. LED ring acts as platter rim glow with proportional warm amber. Premium and audiophile — connects electric light to analog audio warmth. The aesthetic is "luxury audio instrument" (Bang & Olufsen, Technics SL-1200), NOT nightclub or media player. Uses abstract track bands instead of thin grooves to avoid aliasing/Moire at 240×240.

### Concept 20: Eclipse Corona (FINAL CONCEPT)	
Item	Status
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ITEM	STATUS
Compile (ESPHome 2026.5.0)	PASSED (0 errors, 165.61s fresh build)
Physical validation	NOT TESTED

Concept 20 (Eclipse Corona) is the FINAL and MOST PREMIUM concept in the 20-concept UI prototype program. It maps brightness to the intensity and radius of a solar corona radiating from behind a dark moon disk. At 100% brightness, the corona is massive and bright (Solar Maximum — warm amber glow fills the display). At 5%, only a faint, thin sliver of corona is visible (Partial Eclipse). The round display IS the eclipsed sun viewed head-on. A dark "moon disk" (80px solid circle) sits at center, providing a canvas for the percentage readout. Five concentric glow rings (`lv_obj` circles with 15%/25%/40%/55%/75% opacity in warm amber tones from `0xFF8C00` to `0xFFD000`) simulate the radial falloff of the corona. Three asymmetric `lv_arc` streamer segments at different angles (30°-90°, 200°-270°, 340°-20°) break perfect symmetry to create the organic look of real coronal streamers. Page 0 (Power) shows "ECLIPSE" (ON, bright amber corona ring) vs "TOTALITY" (OFF, dark ring). Page 1 (Brightness Hero) shows the full corona stack with streamer arcs and percentage readout on the moon disk. Page 2 (Presets) shows four miniature eclipse zones arranged in a cross pattern: Solar Maximum (Warm White 80%), Golden Corona (Soft Amber 60%), Full Totality (Neutral White 90%), Partial Eclipse (Low Nightlight 15%). LED ring acts as the outermost corona halo, projecting warm amber light onto the physical wall — extending the eclipse beyond the screen boundary. The aesthetic is "captured cosmic wonder" — cinematic, premium, calm. NOT sci-fi, NOT space wallpaper, NOT solar system toy. Differentiated from Concept 13 (Lunar Phase) by showing radial GLOW around a dark disk rather than illuminated moon surface. Differentiated from Concept 17 (Iris Aperture) by using organic soft glow rather than mechanical metallic blades. This is the concept most likely to make VelaDial iconic — the device people photograph and share. 99-thread ultra-wide research (546KB). The capstone of the entire 20-concept program.

Multi-Theme Firmware Engine

Item	Status
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Compile (ESPHome 2026.5.0)	PASSED (0 errors, PR #56 merged)

ITEM	STATUS
Physical validation	NOT TESTED

The multi-theme firmware engine (PR #56, merged) integrates all 20 concept themes into a single compile-passing firmware candidate ([esphome/door_side_rotary.yaml](#)). A hardware validation package exists at [docs/validation/multi_theme_hardware_validation_plan.md](#) with a fillable test results template at [docs/validation/multi_theme_test_results_template.md](#) . All physical test results remain PENDING until Hardik performs testing on real hardware.

Document control

Version: 0.28 — Added multi-theme firmware engine entry; hardware validation package created; all physical results remain NOT TESTED / PENDING. **Owner approval required:** Yes, before lifting the Step 15B gate.

Next phase after sign-off: Door-side sensor validation ([docs/13_Firmware_Prep_Validation_Plan.md](#) §3.C).